	Document Title: Suspension Processing Traveler	Revision: A
	Document #: TR-C.SLQ001	Effective Date: 23Sep2022

General Production Information

Work Order #:	2302-01	**NCMR #:**	N/A
Part Number:	C.SLQ001	**Part Revision:**	B
Lot #:	2302-01-1	**Manufacture Date / Expiration Date:**	Mfg date: 02/22/2023 Exp date: 02/22/2023
Build Purpose:	Production		
Additional Build Info:			

Material Traceability

Part Number	Description	Part Revision	Lot / Batch #	Expiration Date	Qty. Obtained	Obtained By: (Initial/Date)
M.SLQ001	Polymeric Dispersion Solution, Unpurified, CHEMPSBTCI010	B	CRYLK-UX	30 Apr 2023	10.24	GR 02/22/2023

Equipment Traceability

Part Number	Description	Equipment ID	Calibration Due Date
E.SLQ006	Diafiltration System, KrosFlo KTF-2000	ST-005	11/01/2024
E.SLQ010	Floor Scale, Concentrate Reactor	ST-006	09/07/2023
E.SLQ012	Floor Scale, Suspension	ST-007	04/09/2023

Build Record

Step	Procedure	Step Description	Status / Results	Performed By: (Initial/Date)
1	QM.SLQ049	Line clearance completed	☑ Yes ☐ No ☐ N/A	GR 02-22-23
2	MP-C.SLQ001	Drain liquid in diafiltration system/tubing into waste tank.	☑ Yes ☐ No	GR 02-22-23

Step	Procedure	Step Description	Status / Results	Performed By: (Initial/Date)
3	MP-C.SLQ001	Connect Retantate Lines and Feed Lines to the appropriate pumps. Route lines through the appropriate membrane modules, flow meters, and backpressure valves.	☑ Yes ☐ No	GA 02-22-23
4	MP-C.SLQ001	Place Permeate Line into Peristaltic Pump; change Valve 1 to 'permeate line' position.	☑ Yes ☐ No	GA 02-22-23
5	MP-C.SLQ001	Tare Concentrate Reactor Floor Scale. Fill Concentrate Reactor with ~30kg of high purity water (S.SLQ001).	☑ Floor scale tared Floor scale reading: 30.0	GA 02-22-23
6	MP-C.SLQ001	'Feed (remaining)' value on system software is within ±0.2kg of floor scale reading.	'Feed (Remaining)' value: 30.0 ☑ Within ±0.2kg of floor scale reading	GA 02-22-23
7	MP-C.SLQ001	Manually run Pump 01 and Pump 02 until membrane modules and retentate lines are filled.	☑ Yes ☐ No	GA 02-22-23
8	MP-C.SLQ001	Confirm no leaks are observed in all connections.	☑ No leaks observed	GA 02-22-23
9	MP-C.SLQ001	Fill Concentrate Reactor until scale reads ~30kg.	Floor scale reading: 30.0	GA 02-22-23
10	MP-C.SLQ001	Set parameters on main page of system software. • Starting Volume: 30L • DV Setpoint: 2.0 DV	☑ Parameters set	
11	MP-C.SLQ001	Click START PROCESS button to fill membrane modules with water. After membrane modules are filled with water, ensure Permeate Valve is set to vertical position and open the four clamps on permeate line.	☑ Membrane modules filled with water ☑ Permeate Valve set to vertical position ☑ Open four clamps on permeate line	
12	MP-C.SLQ001	Click Stop button when 'Perm. Total' value reads 60L on system software.	'Perm. Total' value: 60.4	GA 02-22-23
13	MP-C.SLQ001	Mount Pneumatic Drum Stirrer into holder and turn on.	☑ Yes ☐ No	

Step	Procedure	Step Description	Status / Results	Performed By: (Initial/Date)
14	MP-C.SLQ001	Obtain container(s) of Unpurified Polymeric Dispersion Solution (M.SLQ001). Record traceability information in the Material Traceability table above.	☑ Container(s) obtained ☑ Material traceability recorded	GR 02-22-23
15	MP-C.SLQ001	Record volume of container(s).	Volume of container(s): 10.24	
16	MP-C.SLQ001	Fill Concentrate Reactor with high purity water until the Designated Fill Value (DFV) equals the volume of container(s).	☑ Yes ☐ No Designated Fill Value (DFV): 32.1	GR 02-22-23
17	MP-C.SLQ001	Calculate the Designated Operating Value (DOV). DOV = 10 / 3 * DFV.	Designated Operating Value (DOV): 107.0	GR 02-22-23
18	MP-C.SLQ001	Slowly pour contents of Unpurified Polymeric Dispersion Solution container(s) into concentrate reactor. Ensure no gelation develops.	☑ Yes ☐ No	
19	MP-C.SLQ001	Fill concentrate reactor with high purity water until DOV is reached (within ±2kg). Record floor scale and system software 'Feed (Remaining)' value. Verify values are within ±2kg of DOV and each other.	Floor scale reading: 107.2 "Feed (Remaining)" value: 107.2 ☑ Values are within ±2kg of DOV and each other	GR 02-22-23
20	MP-C.SLQ001	Set parameters on main page of system software. • Starting Volume: Designated Operating Value • DV Setpoint: 10 DV	☑ Parameters set	
21	MP-C.SLQ001	Click START PROCESS button. Verify Pump 01 and Pump 02 are running	☑ Pump 01 and Pump 02 running	
22	MP-C.SLQ001	Run process until "Perm. Total" value reaches 10 times the value of the Designated Fill Value (see Step 17 above).	"Perm. Total" value: 1084.2	
23	MP-C.SLQ001	Record weight of product tank (C.SLQ003) without the lid.	Product tank weight: 10.2 kg	GR 02-22-23

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Step	Procedure	Step Description	Status / Results	Performed By: (Initial/Date)
24	MP-C.SLQ001	Transfer Suspension to plastic drum using the peristaltic pump via the product line (C.SLQ003).	☑ Yes ☐ No	GR 02-22-23
25	MP-C.SLQ001	Take sample of suspension in 50mL tubes at the beginning, middle, and end of the transfer. Label each tube Aliquot A, Aliquot B, and Aliquot C, respectively.	☑ Yes ☐ No	GR 02-22-23
26	MP-C.SLQ001	Place clean containers below Pump 01 and Pump 02. Disconnect retentate lines from pumps and allow suspension to drain into containers. Pour contents of the containers into the product tank.	☑ Yes ☐ No	GR 02-22-23
27	MP-C.SLQ001	Record the initial weight of the suspension. Assumption: The initial weight (in kg) of the suspension is equal to the initial volume (in L).	Initial Suspension Weight: 111.6 kg	GR 02-22-23
28	MP-C.SLQ001	Seal product tank.	☑ Yes ☐ No	GR 02-22-23
29	MP-C.SLQ001	Create product label for Suspension (L.SLQ005) indicating lot number and expiration date of suspension. Apply label to product tank.	☑ Yes ☐ No	
30	MP-C.SLQ001	Place product tank in quarantine pending QC analysis of suspension.	☑ Yes ☐ No	
31	MP-C.SLQ001	***Prepare KTF-2000 System for storage*** Fill concentrate reactor with ~30kg of high purity water and manually start running Pump 01 and Pump 02. Allow Membrane Modules 1 and 2 and Retentate Lines 1A, 1B, 2A, and 2B are filled. Allow pumps to run for an additional 10 minutes.	☑ Yes ☐ No	
32	MP-C.SLQ001	Switch Permeate Line Valve to horizontal (closed) position and Waste Line Valve to vertical (open) position. Place Waste Line through peristaltic pump and into the concentrate reactor. Manually run the peristaltic pump to fully drain the contents of the concentrate reactor into the waste tank.	☑ Yes ☐ No	
33	MP-C.SLQ001	Create Storage Solution. Combine 20L of high purity water + 1.5mL of bleach solution (S.SLQ007).	☑ Yes ☐ No	

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Step	Procedure	Step Description	Status / Results	Performed By: (Initial/Date)
34	MP-C.SLQ001	Pour Storage Solution into Concentrate Reactor. Close 4 Permeate Line Clamps and switch Permeate Line Valve to horizontal (closed) position and Waste Line Valve to vertical (open) position.	☑ Yes ☐ No	GR 02-22-23
35	MP-C.SLQ001	Manually run Pump 01 and Pump 02 until Membrane Modules 1 and 2 and Retentate Lines 1A, 1B, 2A, and 2B are filled.	☑ Yes ☐ No	
36	MP-C.SLQ001	Once filled, stop Pump 01 and Pump 02 and close the ball valves for Retentate Lines 1B and 2B at the Concentrate Reactor. Allow 5 minutes to elapse for settling and degassing. Ensure the Storage Solution has filled Membrane Modules 1 and 2 entirely. Storage Solution should at least fill Retentate Lines 1B and 2B up to the respective line's Flow Meter.	☑ Yes ☐ No	
37	MP-C.SLQ001	Wait 5 minutes to ensure the Membrane Modules are full and stable.	☑ Yes ☐ No	
38	MP-C.SLQ001	Manually run the Peristaltic Pump to fully drain the contents of the Concentrate Reactor into the Waste Tank.	☑ Yes ☐ No	
39	MP-C.SLQ001	Power down Pump 03, Pump 04, concentrate reactor scale, suspension scale, and KTF system software.	☑ Yes ☐ No	
40	MP-C.SLQ001	***Prepare the high purity water tank for storage*** Take a peristaltic pump with sterilized tubing to pump out any residual high purity water from the tank.	☑ Yes ☐ No	
41	MP-C.SLQ001	Allow high purity tank to dry for 24 hours before securing lid to the tank.	☑ Yes ☐ No	
42	MP-C.SLQ001	Clean containers using in the manufacturing process. Allow containers to dry and store after they have dried.	☑ Yes ☐ No	
43	QC-C.SLQ001	Execution of QC analysis of suspension samples. QC analysis report attached to DHR.	☑ Yes ☐ No	
44	MP-C.SLQ001	Attach Suspension Quality Control Report and COA to DHR.	☑ Yes ☐ No	GR 02-22-23
45	MP-C.SLQ001	Test results meet acceptance criteria.	☑ Yes ☐ No	GR 02-22-23

Reviewed By (Signature and Date): Bryan Wong 27 Mar 2023 Batch/Lot No.: 2302-01-1

Step	Procedure	Step Description	Status / Results	Performed By: (Initial/Date)
46	MP-C.SLQ001	***Dilution Procedure*** Calculate the Average Suspension Sample Concentration (g/L) by taking the average 'Residue Net Weight' of six (6) freeze dried suspension samples based off the 10 mL sample.	Average Suspension Sample Concentration: 12.787 g/L	GR 02-22-23
47	MP-C.SLQ001	Calculate the volume (L) of high purity water to be added to the product tank. Volume = [Average Suspension Sample Concentration (g/L) * Initial Suspension Volume (L)]/[10 (g/L)] – Initial Suspension Volume (L)	Volume of high purity water to be added to product tank: 28.3 L	GR 02-22-23
48	MP-C.SLQ001	Tare suspension floor scale and place product tank onto floor scale.	☑ Yes ☐ No	GR 02-22-23
49	MP-C.SLQ001	Add volume of high purity water (from Step 35) to the product tank.	☑ Yes ☐ No	GR 02-22-23
50	MP-C.SLQ001	Record the weight of product tank.	139.8 kg	GR 02-22-23
51	MP-C.SLQ001	Calculate weight of suspension.	129.6 kg	GR 02-22-23
52	MP-C.SLQ001	Update product label with weight of suspension.	☑ Yes ☐ No	GR 02-22-23

Comments/Additional Notes:

Part Description: Suspension

Part Number: C.SLQ001

Revision: B

Amount: 129.6kg

Lot Number: 2302-01-1

Expiration 22FEB2024

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QUARANTINE